

Knowledge Representation & Reasoning in AI

Part 1
Semantic Web

Part 2
RDF & OWL

Part 3
SPARQL

Part 4
Probability
& Bayes

Part 5
Markov &
Fuzzy

Lecture 4 Dr. Husnain Ashfaq

PART 1

The Semantic Web

Teaching computers to UNDERSTAND information — not just display it

What's Wrong With the Regular Web?

❌ Old Web — Word Matching

Search: "father of Albert Einstein"

Results:

- Finds pages with word 'father'
- Finds pages with word 'Einstein'
- No MEANING understood
- Just pattern matching!

**Computer sees WORDS
NOT MEANING**

✅ Semantic Web — Understands Meaning

Query: "father of Albert Einstein"

Albert Einstein

IS A

Person

Hermann Einstein

IS A

Person

Hermann Einstein

FATHER OF

Albert

**Answer: Hermann Einstein
(father = family relationship)**

Three Generations of the Web

Web 1.0



(1990s) *The Library*

Web 2.0



(2004+) *Library + Café*

Web 3.0



(Today) *Smart Library*

- Read-only pages
- Static HTML docs
- No user interaction
- Uni homepage 1999

- Read AND Write
- Social media
- YouTube, Wikipedia
- Facebook, Twitter

- Machines understand
- Facts linked by meaning
- Direct answers
- Google Knowledge Graph

Semantic Web Technology Stack — Layer Cake

Built in layers — each layer depends on the one below. Higher = more powerful.

Layer 7: TRUST

Can I believe this data?

Layer 6: RULES (SWRL)

IF...THEN logic statements

Layer 5: OWL

Defines what categories MEAN

← We study this

Layer 4: RDFS

Basic types & category schemas

Layer 3: RDF

Facts stored as triples

← We study this

Layer 2: URI

Every thing gets a unique web ID

← We study this

Layer 1: XML / Unicode

The alphabet & grammar of the web

▲ Build from BOTTOM UP

Real World: Google Knowledge Graph



500 Billion

Facts in Google's
Knowledge Graph



Linked Data

Facts connected
across web pages



Direct Answers

No website click
needed — AI answers

Eiffel Tower



Location

Paris, France



Height

330 meters



Built

1887–1889



Architect

Gustave Eiffel



Visitors/yr

~7 million



Type

Iron lattice tower

← Semantic Web in action!

Quiz — The Semantic Web

Think before checking answers!

Q1

Saad builds a hospital directory showing names, addresses and phone numbers in an HTML table. His friend says it's already a Semantic Web application. Who is right and why? What is missing for it to be truly 'semantic'?

Q2

A healthcare app answers: 'Which cardiologist in Lahore is available on Saturday?' — returning Dr. Azhar's profile directly without showing a list of websites. Which web generation is this? What TWO technologies make it possible?

Q3

Facebook lets users post and comment. Wikipedia lets anyone edit articles. Which web generation are these? What specific change would be needed for them to qualify as Web 3.0?



Quiz Answers – The Semantic Web



Model Answers

A1

Saad is RIGHT. HTML only tells the browser HOW to display info (layout, fonts). The Semantic Web requires machines to understand WHAT the info means — 'this string is a hospital name', 'this is a geographic location'. Missing: RDF/OWL markup that adds semantic meaning machines can process.

A2

Web 3.0. Technologies: (1) RDF stores doctor profiles as linked triples — connecting doctor, specialisation, location, availability. (2) SPARQL or a Knowledge Graph query engine retrieves the answer by pattern-matching triples rather than keyword-searching web pages.

A3

Both are Web 2.0 — users create and share content interactively. For Web 3.0 they need: semantic markup (RDF tags) on the data so machines can understand entities and relationships. A computer cannot 'read' a Wikipedia paragraph and truly know the relationships between entities mentioned in it.

Activity — Classify Each Example Into a Web Generation

 Individual Activity (8 min) — Write Web 1.0, 2.0 or 3.0 for each and give ONE reason.

1

A static university homepage from 1998 listing course names

—

2

Google answering 'Who invented the telephone?' with Alexander Graham Bell directly

—

3

A YouTube video with likes, comments, and user uploads

—

4

Siri telling you 'The nearest open pharmacy is 0.3 km away'

—

5

A read-only product catalogue on a retailer website (2001)

—

6

WhatsApp showing you real-time messages from friends

—

7

Wikidata returning all properties of 'Lahore' as structured RDF triples

—

8

A news site where readers vote and rank stories (Reddit-style)

—



Activity Solution – Web Generation Classification



Model Answers

Web 1.0

1. Static 1998 university homepage

Read-only, no interaction

Web 3.0

2. Google answers 'Who invented phone?'

Machine understands meaning

Web 2.0

3. YouTube with likes & comments

User-generated, interactive

Web 3.0

4. Siri finds nearest open pharmacy

Knowledge graph + context

Web 1.0

5. Read-only 2001 product catalogue

Static display only

Web 2.0

6. WhatsApp real-time messages

Dynamic, user-to-user

Web 3.0

7. Wikidata RDF triples for Lahore

Structured semantic data

Web 2.0

8. Reddit-style voting on stories

Interactive participation

PART 2

RDF & OWL

Storing facts as triples • Defining rules with ontologies

RDF — Facts as Three-Part Statements (Triples)

Every fact in RDF = one TRIPLE: Subject → Predicate → Object (just like grammar!)

English:

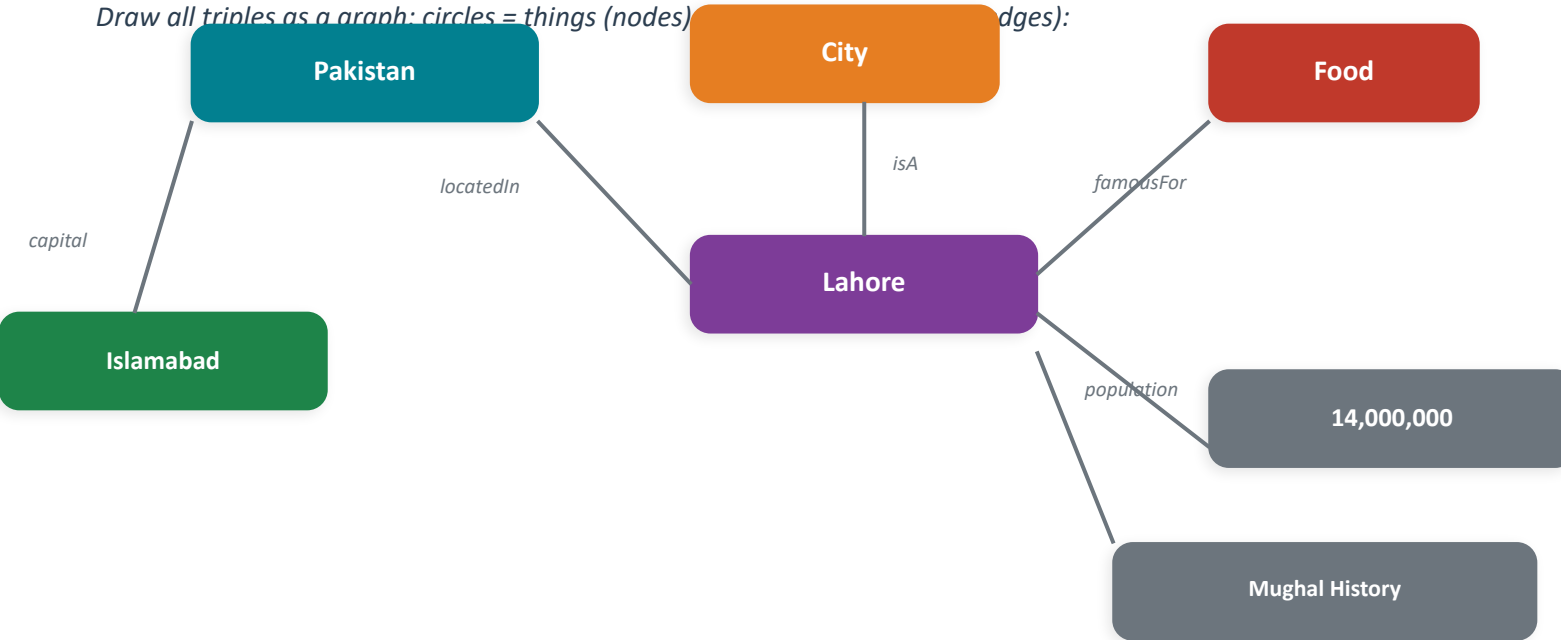
"Ali studies Computer Science"



Subject	Predicate	Object
Lahore	isA	City
Lahore	locatedIn	Pakistan
Lahore	population	14,000,000
Pakistan	capital	Islamabad
Pakistan	isA	Country

RDF Knowledge Graph — Nodes and Arrows

Draw all triples as a graph: circles = things (nodes) edges:



URI = Unique ID for every thing (like a CNIC for web resources). `<http://dbpedia.org/resource/Lahore>` is Lahore's unique URI — the word 'Lahore' alone could mean anything, but the URI makes it globally unambiguous.

Writing RDF — Turtle Syntax: 4 Rules to Remember

Turtle (.ttl) is the most readable RDF format. Four punctuation symbols do all the work:

. — PERIOD

Ends one complete triple statement

```
Lahore isA City .
```

Note:
Like a full stop in English

, — COMMA

Same subject AND predicate — new object

```
Lahore famousFor  
Food , History .
```

Note:
*Multiple values, one
predicate*

; — SEMICOLON

Same subject — new predicate (no repeat)

```
Lahore isA City ;  
population 14M .
```

Note:
Avoids repeating 'Lahore'

@prefix — PREFIX

Shortcut for a long web address

```
@prefix ex:  
<http://example.org/> .
```

Note:
Saves typing long URIs

Activity — Write RDF Triples for FAST University



Pair Activity (10 min) — Fill every blank cell. Then draw the RDF graph on paper.

#	Fact (plain English)	Subject	Predicate	Object
1	FAST Lahore is a University	FAST_Lahore	?	?
2	FAST is located in Lahore	?	locatedIn	?
3	FAST was founded in 1985	?	?	1985
4	FAST offers Computer Science	FAST_Lahore	?	?
5	FAST offers AI program	?	offers	?
6	Enrollment is 5000 students	?	?	5000

BONUS: Draw the RDF graph — put FAST_Lahore in the centre. Draw 6 arrows (one per triple) to the object nodes. Label each arrow with the predicate.

✓ Activity Solution – FAST University RDF Triples

✓ Model Answers

#	Fact	Subject	Predicate	Object
1	FAST Lahore is a University	FAST_Lahore	rdf:type	University
2	FAST is located in Lahore	FAST_Lahore	locatedIn	Lahore
3	FAST was founded in 1985	FAST_Lahore	foundedYear	1985
4	FAST offers Computer Science	FAST_Lahore	offers	ComputerScience
5	FAST offers AI program	FAST_Lahore	offers	AI
6	Enrollment is 5000 students	FAST_Lahore	enrollment	5000

University

CS

FAST_Lahore

Lahore

AI

Key: FAST_Lahore is the central node in ALL 6 triples. Two 'offers' predicates create two separate arrows → CS and AI.

OWL — Rules About What Knowledge MEANS

RDF stores FACTS. OWL stores the RULES about those facts — the instruction manual.

RDF = Spreadsheet of Facts

Fido → isA → Dog
Fido → hasOwner → Ali
Ball → color → Red

(Just raw facts — no meaning attached to 'Dog')

OWL = Instruction Manual (Rules)

Dog is a subtype of Animal
Dog and Cat CANNOT overlap
Every Dog has exactly ONE breed
If X parentOf Y, Y parentOf Z → X grandParentOf Z

Ontology

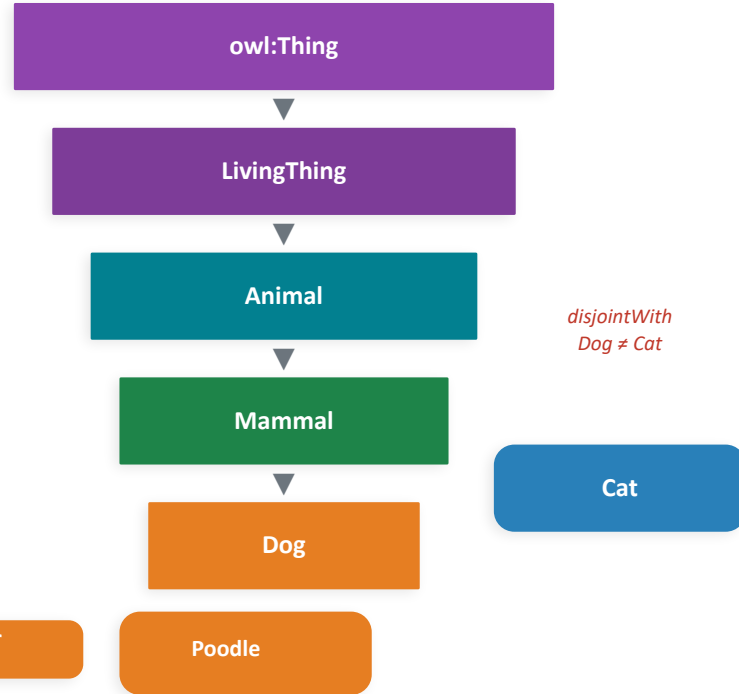
A formal description of a domain: what THINGS exist, what TYPES they belong to, and what RELATIONSHIPS connect them. Like a detailed dictionary + rulebook.
Real example: SNOMED CT medical ontology has 350,000+ concepts used by NHS (UK), Australia, and hospitals worldwide.



OWL in Healthcare: hierarchy Disease → InfectiousDisease → ViralInfection → COVID-19. A drug treating 'ViralInfection' is AUTOMATICALLY known to help COVID-19 — the OWL reasoner infers it from the hierarchy without anyone adding the link manually!

OWL Class Hierarchy – The 'Is-A' Pyramid

Classes in OWL are organised in a hierarchy — what is true of the parent is inherited by children:



rdfs:subClassOf

`Dog subClassOf Mammal`

Every Dog IS a Mammal — hierarchy + inheritance

owl:disjointWith

`Dog disjointWith Cat`

Nothing can be BOTH — mutual exclusion

owl:equivalentClass

`Car ≡ Automobile`

Two class names mean the SAME thing

owl:unionOf

`Parent = Father OR Mother`

Member of at least one of the classes

OWL Properties — How Things Relate

Properties describe how two things relate. Two kinds — and three special behaviours:

Object Property owl:ObjectProperty

Links a THING to another THING

Subject	Predicate	Object
Ali (Person)	worksAt	FAST (Organisation)

Datatype Property owl:DatatypeProperty

Links a THING to a VALUE

Subject	Predicate	Value
Ali (Person)	age	25 (Integer)



Transitive

Chains automatically

isAncestorOf: Ali→Bilal, Bilal→Carla ∴ Ali→Carla (auto-inferred)



Symmetric

Works both ways

isSiblingOf: Sara→Ahmed ∴ Ahmed→Sara (auto-added)



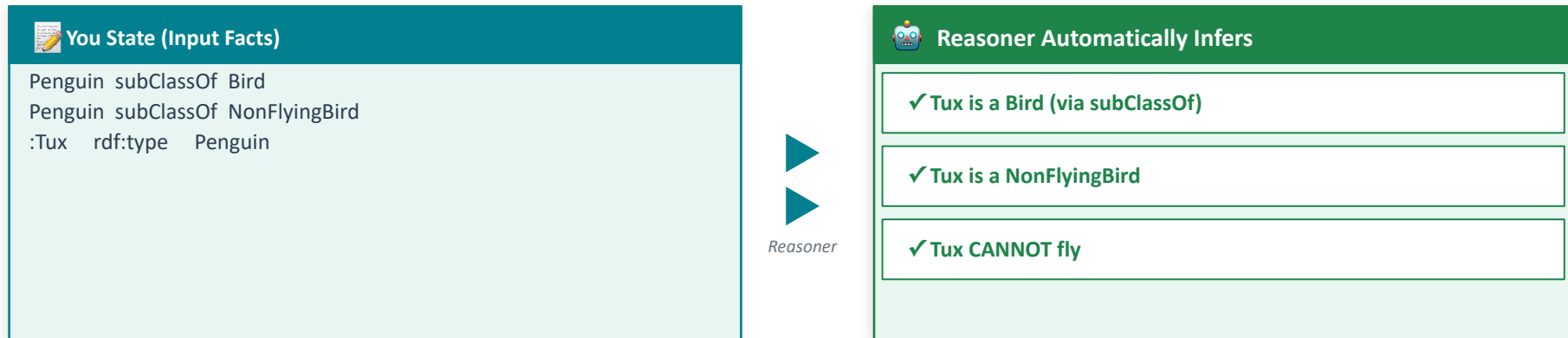
Functional

One value only

hasBiologicalMother: Ali has EXACTLY ONE — a second value = error

OWL Automated Reasoning – The Computer Infers New Facts

Give an OWL ontology + facts → the reasoner automatically infers new truths and catches errors:



What Happens With a Contradiction?

Add: ':Tux rdf:type :Eagle' → Eagle is a FlyingBird. Tux is ALREADY NonFlyingBird. These are DISJOINT → INCONSISTENCY! The reasoner flags this error automatically — why hospitals use OWL to catch contradictions in patient records before they cause harm.



SNOMED CT: 350,000+ medical OWL concepts. NHS London + Sydney share records. If Drug X treats ViralInfections, OWL automatically knows it may help COVID-19 (subclass). No one needs to add that link manually!

Quiz — OWL Ontology

Think before checking answers!

Q1

Hospital OWL says: `Surgeon subClassOf Doctor`, `Doctor subClassOf HealthcareWorker`. Record states `':Ahmed rdf:type :Surgeon'`. Without adding any more triples, what TWO facts can the OWL reasoner automatically infer about Ahmed?

Q2

A data entry says: `':Robot99 rdf:type :Student'` AND `':Robot99 rdf:type :Teacher'`. The OWL says: `Student owl:disjointWith Teacher`. What does the reasoner do? Is it an error or a valid state? Why is this actually USEFUL?

Q3

Design a mini university OWL ontology: (a) ONE `subClassOf` relationship, (b) ONE `disjointWith` relationship, (c) ONE `TransitiveProperty` — all related to university. Explain what each rule prevents or enables the reasoner to do.



Quiz Answers — OWL Ontology



Model Answers

A1

(1) Ahmed is a Doctor — via Surgeon subClassOf Doctor. (2) Ahmed is a HealthcareWorker — via Doctor subClassOf HealthcareWorker. The OWL reasoner inherits these facts automatically through the class chain. This is called 'entailment' — the hierarchy propagates upward without you writing extra triples.

A2

INCONSISTENCY ERROR. Disjoint means nothing can belong to both. Reasoner flags a contradiction. This is a FEATURE — it catches data entry errors (perhaps Robot99 is a Teaching Assistant, not a full Student AND Teacher). Automatic error detection is the whole point of OWL's disjointWith constraint.

A3

(a) subClassOf: GraduateStudent subClassOf Student — all graduate students ARE students, inheriting all student properties automatically. (b) disjointWith: Student disjointWith StaffMember — one person cannot be simultaneously enrolled and employed (simplification). (c) TransitiveProperty: 'isPrerequisiteOf' — if A prereq B, B prereq C, then A is automatically prereq C, building the full chain.

Activity – OWL Reasoning: What Can Be Inferred?

For each scenario, write what the OWL reasoner can infer or flag.

1. Zoo Hierarchy

- `Lion subClassOf Mammal`
- `Mammal subClassOf Animal`
- `Animal subClassOf LivingThing`

? What 3 facts are automatically inferred about Simba?

2. Disjoint Classes

- `Car owl:disjointWith Bicycle`
- `:Tesla rdf:type :Car`
- `:Tesla rdf:type :Bicycle`

? What does the reasoner do? Is Tesla valid?

3. Transitive Property

- `isPartOf is TransitiveProperty`
- `Wheel isPartOf Engine`
- `Engine isPartOf Car`

? What triple is automatically inferred?

4. Symmetric Property

- `isMarriedTo is SymmetricProperty`
- `:Ali isMarriedTo :Fatima`

? What new triple does the reasoner add?

✓ Activity Solution — OWL Reasoning Chains

✓ Model Answers

1. Zoo

Simba is a Mammal (Lion→Mammal), Simba is an Animal (Mammal→Animal), Simba is a LivingThing (Animal→LivingThing). Three inferences from ONE stated fact — the entire chain propagates automatically.

2. Disjoint

INCONSISTENCY ERROR. Tesla cannot be both Car and Bicycle — disjoint classes. Reasoner flags the contradiction and marks the ontology inconsistent. The incorrect triple must be removed or corrected.

3. Transitive

Wheel isPartOf Car — inferred automatically! Transitivity chains: A isPartOf B, B isPartOf C → A isPartOf C. The reasoner adds this triple without anyone writing it explicitly.

4. Symmetric

:Fatima isMarriedTo :Ali is automatically inferred. Symmetric property works both ways — no need to state it twice. The reasoner adds the reverse triple automatically to maintain consistency.

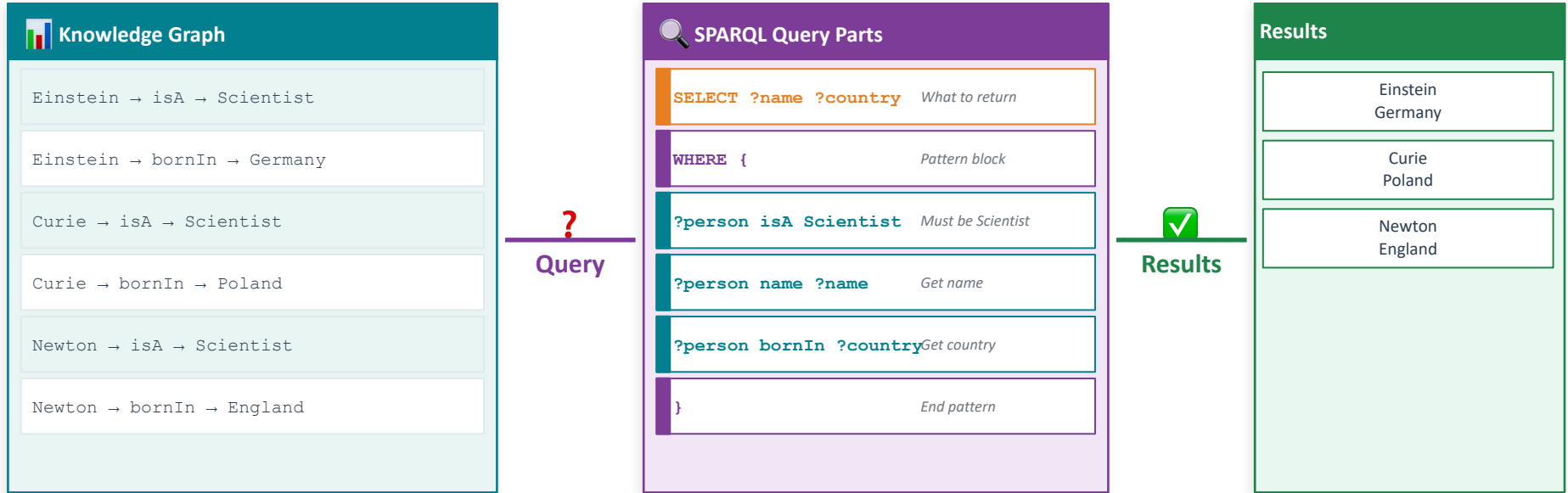
PART 3

SPARQL — Query Language

Pattern matching on knowledge graphs — asking questions

SPARQL — How Pattern Matching Works

SPARQL finds triples matching a pattern — question marks (?) are blanks that get filled in:



The '?' symbol = VARIABLE (a blank to fill in). ?person gets filled with Einstein, Curie, Newton... ?name gets their names. SPARQL finds ALL combinations that match the pattern simultaneously!

SPARQL — 4 Query Types at a Glance



SELECT

Retrieve data from the graph

Example: Find all Action movies → returns a table

Returns: A TABLE of matching results



ASK

Get a TRUE or FALSE answer only

Example: Is Inception a SciFi film? → TRUE or FALSE

Returns: TRUE or FALSE only



CONSTRUCT

BUILD new triples from existing data

Example: Combine first+last name → create fullName triple

Returns: A NEW set of triples



DESCRIBE

Return ALL facts about one resource

Example: Describe :Islamabad → everything known

Returns: ALL triples about it

SPARQL — FILTER and OPTIONAL Keywords

FILTER narrows results. OPTIONAL includes rows even when some data is missing:

FILTER — Like a Sieve

All movies (before FILTER):

Movie	Year
Inception	2010
Gladiator	2000
Interstellar	2014
The Notebook	2004

FILTER (year > 2005)

After FILTER (year > 2005):

Movie	Year
Inception	2010
Interstellar	2014

OPTIONAL — Like a Left Join

Find all students AND their phone
(include students even without phone):

Student	Phone
Ali	0321-1234567
Sara	(empty — no phone)
Ahmed	0300-9876543
Zara	(empty — no phone)

Without OPTIONAL:

Sara & Zara excluded (no phone = no row)

With OPTIONAL:

ALL students appear — phone blank if missing

Activity — SPARQL Movie Database Queries

Choose the correct query type and write the WHERE pattern.

Knowledge Graph — Movie Database

Inception

Nolan | 2010 | SciFi | 8.8

Interstellar

Nolan | 2014 | SciFi | 8.6

Gladiator

Scott | 2000 | Action | 8.5

Q1

SELECT

Find all movies directed by Christopher Nolan

Pattern: `?movie :director :Nolan`

Q2

FILTER

Find movies with a rating ABOVE 8.6

Use FILTER with a numerical condition

Q3

ASK

Is there at least ONE SciFi movie in the database?

ASK returns only TRUE or FALSE

Q4

COUNT

How many movies has each director made? Sort most to least.

Use `COUNT(?movie) AS ?total, GROUP BY ?director, ORDER BY DESC`

Activity Solution — SPARQL Movie Queries



Model Answers

Q1 — Nolan movies [SELECT]

```
SELECT ?movie
```

```
WHERE { ?movie :director :Nolan . }
```

→ Inception, Interstellar

Q2 — Rating > 8.6 [FILTER]

```
SELECT ?movie ?rating
```

```
WHERE { ?movie :rating ?rating .
```

```
  FILTER(?rating > 8.6) }
```

→ Inception (8.8)

Q3 — Any SciFi? [ASK]

```
ASK {
```

```
  ?m :genre :SciFi . }
```

→ TRUE

Q4 — Director tallies [COUNT]

```
SELECT ?dir (COUNT(?movie) AS ?n)
```

```
WHERE { ?movie :director ?dir . }
```

```
GROUP BY ?dir ORDER BY DESC(?n)
```

→ Nolan: 2, Scott: 1

Quiz — SPARQL Queries

Think before checking answers!

Q1

A university knowledge graph has triples: `:Ali isA :Student`, `:Ali enrolled :CS`, `:Ali cgpa '3.8'`. Write the WHERE pattern (in English or SPARQL) to find all CS students with CGPA above 3.5. What keyword adds the numerical condition?

Q2

You want to check if lecturer `:DrKhan` has ever taught `:AI101`. You only need YES or NO — which of the 4 query types do you use? Why is SELECT wrong here? Write the WHERE pattern.

Q3

A hospital query needs all patients AND their blood type, but MUST include patients even if blood type is not recorded. Which SPARQL keyword do you need? Draw a 3-row result table showing what a patient with NO blood type looks like in the results.

Quiz Answers — SPARQL Queries



Model Answers

A1

WHERE pattern needs 3 triple patterns (Student + enrolled CS + cgpa) AND a FILTER keyword for the numerical condition: { ?student isA :Student . ?student enrolled :CS . ?student cgpa ?cgpa . FILTER(?cgpa > 3.5) }. All three triple patterns are AND-ed together — must match all simultaneously.

A2

Use ASK query type. SELECT is wrong because it returns a table of results — you don't need data, just existence. ASK returns TRUE if the triple ':DrKhan :taught :AI101' exists in the graph, FALSE otherwise. WHERE pattern: { :DrKhan :taught :AI101 . }. One-line pattern.

A3

Use OPTIONAL keyword. Result table: Row 1: Ali | O+ (has blood type). Row 2: Sara | (empty/null — no blood type recorded). Row 3: Ahmed | A-. Without OPTIONAL, Sara's row would be completely absent from results. OPTIONAL keeps her row and leaves the blood type cell empty — just like a SQL LEFT JOIN.

PART 4

Probability & Bayes' Theorem

Handling uncertainty — updating beliefs when new evidence arrives

Why Classical Logic Fails in the Real World

Classical logic is a light switch — ON or OFF. Real AI needs a DIMMER (0 to 1):

Missing Info

Ali's car not in database

❌ Logic: No car

✅ Reality: Maybe just not recorded yet!

Uncertain Data

Thermometer: 36.9°C ($\pm 0.5^\circ$ error)

❌ Logic: No fever

✅ Reality: Could actually be 37.3°C!

Conflicting Rules

Birds fly. Penguins are birds.

❌ Logic: Penguin can fly → CONTRADICTION

✅ Reality: Need to handle exceptions

Vague Words

Is Lahore a 'big' city?

❌ Logic: Needs exact yes/no

✅ Reality: 14M = big vs UAE, small vs Tokyo

World Changes

President X is in office

❌ Logic: No concept of time

✅ Reality: Facts expire and change

Why Classical Logic Fails

What is Classical Logic?

Classical logic works like a light switch — only ON (True) or OFF (False). There is no middle ground. Either something IS true, or it is NOT true.

The 5 Problems with Classical Logic in Real Life:

- 1) Missing Info: Ali's car not in database → Logic says NO CAR. Reality: maybe just not recorded yet!
- 2) Uncertain Data: Thermometer shows 36.9°C ($\pm 0.5^\circ$ error) → Logic says NO FEVER. Reality: could be 37.3°C!
- 3) Conflicting Rules: Birds fly. Penguins are birds. Logic says penguin can fly → CONTRADICTION!
- 4) Vague Words: Is Lahore a 'big' city? Logic needs exact yes/no. Reality: big vs UAE, small vs Tokyo.
- 5) World Changes: President X is in office. Logic has no concept of time. Reality: facts expire!

The Solution:

We need reasoning that handles DEGREES of truth — not just yes/no. That is why AI uses Probability (0 to 1) and Fuzzy Logic.

Probability — The Three Basic Rules

0.0
Impossible

0.25
Unlikely

0.5
50-50

0.75
Likely

1.0
Certain

Rule 1

Complement — The Opposite

$$P(\text{NOT } A) = 1 - P(A)$$

$$P(\text{Rain})=0.7 \rightarrow P(\text{No Rain}) = 1-0.7 = 0.3$$

Rule 2

AND Rule — Both Must Happen

$$P(A \text{ AND } B) = P(A) \times P(B) \quad [\text{if independent}]$$

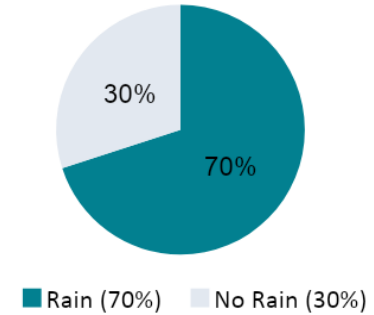
$$P(\text{Heads AND Heads again}) = 0.5 \times 0.5 = 0.25$$

Rule 3

Conditional Probability — Given Something Happened

$$P(A \mid B) = \text{'Probability of A, GIVEN B already happened'}$$

$$P(\text{Ace} \mid \text{card is a Spade}) = 1/13 \text{ — universe shrinks to 13 cards}$$



Probability Rules

Rule 1 — Complement: $P(\text{NOT } A) = 1 - P(A)$

Example: $P(\text{Rain}) = 0.7$, so $P(\text{No Rain}) = 1 - 0.7 = 0.3$

WHY? All probabilities must add up to 1.0. If 70% chance of rain, exactly 30% must be no-rain.

Rule 2 — AND Rule: $P(A \text{ AND } B) = P(A) \times P(B)$ [if independent]

Example: $P(\text{Heads AND Heads again}) = 0.5 \times 0.5 = 0.25$

WHY multiply? For BOTH to happen, coin 1 must be heads (50%) AND THEN coin 2 also heads (50%).

Independent events multiply because each event does not affect the other.

Rule 3 — Conditional Probability: $P(A \mid B) = \text{Prob of } A \text{ GIVEN } B \text{ already happened}$

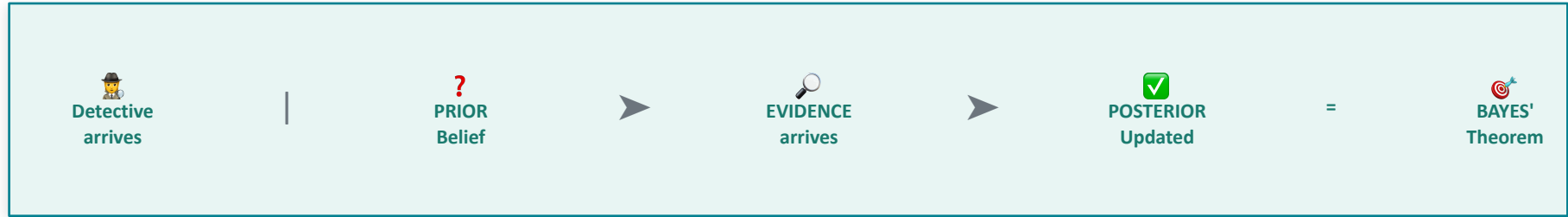
Example: $P(\text{Ace} \mid \text{card is a Spade}) = 1/13$

WHY 1/13? The condition 'card is a Spade' SHRINKS the universe from all 52 cards to only 13 spades.

Of those 13 spades, only 1 is an Ace. So probability = 1/13.

KEY: The $\mid$ symbol always means the sample space shrinks to just the given condition.

Bayes' Theorem — Updating Beliefs With Evidence



$$P(H | E) = \frac{P(E | H) \times P(H)}{P(E)}$$

← Posterior
(THE ANSWER)

P (H)

Prior — Your belief BEFORE seeing any evidence

P (E | H)

Likelihood — How likely is evidence E if H is true?

P (E)

Evidence Prob. — Overall chance of seeing E (all causes)

P (H | E)

Posterior ★ — Belief AFTER evidence — THIS is your answer!

Bayes Theorem — What Each Part Really Means

The Formula: $P(H|E) = P(E|H) \times P(H) / P(E)$

P(H) — Prior Probability:

Your belief BEFORE seeing any evidence. Example: Before any test, you believe $P(\text{Disease}) = 0.01$ — only 1% of the population has this disease.

P(E|H) — Likelihood:

How likely is the evidence IF hypothesis H is true? Example: $P(\text{Positive Test} | \text{Disease}) = 0.95$ means IF the person is sick, the test correctly catches it 95% of the time.

P(E) — Total Evidence Probability:

The overall chance of seeing this evidence from ALL causes. Calculated by: $P(E|H) \times P(H) + P(E|\text{not } H) \times P(\text{not } H)$. This combines the sick AND healthy populations.

P(H|E) — Posterior (THE ANSWER):

Your UPDATED belief AFTER seeing the evidence E. This is what Bayes gives you — a smarter probability that combines your prior belief with the new evidence. Posterior becomes the new Prior for the next evidence!

Bayes' Theorem — Fully Worked: Medical Test Example

Given: Disease affects 1% — $P(D)=0.01$ | Test 95% accurate for sick — $P(+|D)=0.95$ | False alarms 10% — $P(+|no D)=0.10$

Step 1: List What We Know

$$P(\text{Disease}) = 0.01$$

$$P(\text{No Disease}) = 1 - 0.01 = 0.99$$

$$P(+ | \text{Disease}) = 0.95$$

$$P(+ | \text{No Disease}) = 0.10$$

Step 2: Find $P(+)$ — Total Probability

$$P(+) = P(+|D) \times P(D) + P(+|no D) \times P(no D)$$

$$= (0.95 \times 0.01) + (0.10 \times 0.99)$$

$$= 0.0095 + 0.099$$

$$= 0.1085$$

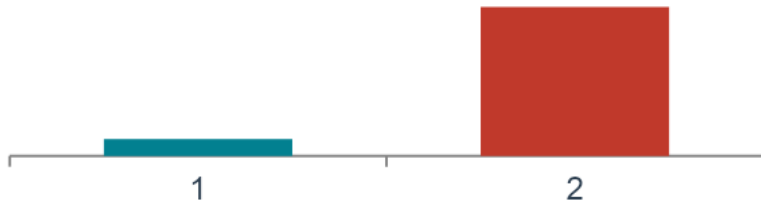
Step 3: Apply Bayes' Formula

$$P(D|+) = P(+|D) \times P(D) / P(+)$$

$$= (0.95 \times 0.01) / 0.1085$$

$$= 0.0095 / 0.1085$$

$$\approx 0.0876 = 8.76\%$$



Surprising Result!

Despite a positive test, only ~9% chance of actually being sick. Why? Disease is RARE (1%), so most positives are FALSE ALARMS from healthy people. This is the BASE RATE effect — prior probability dominates!

Medical Test — Why Only 8.76%?

Your intuition says: Test is 95% accurate → I must be 95% likely sick! This feels right but is **WRONG**.

Imagine 1 crore (10 million) people from Lahore all get tested:

Sick people (1% of 1 crore) = 1,00,000 people are truly sick

Healthy people (99% of 1 crore) = 99,00,000 people are perfectly healthy

Now run the test on ALL of them:

Sick group: Test catches 95% correctly → 95,000 TRUE positives 

Healthy group: 10% false alarm rate → 9,90,000 FALSE positives 

Total positive results = 95,000 + 9,90,000 = 10,85,000 positives

But only 95,000 of them are truly sick!

$95,000 / 10,85,000 = 8.76\%$ — most positives are false alarms!

The KEY INSIGHT — Base Rate Effect:

Healthy people (99,00,000) vastly outnumber sick people (1,00,000). Even a small 10% false alarm rate on that HUGE healthy group creates far more false positives than real cases from the tiny sick group. The rarity of the disease **DOMINATES** everything.

Activity — Apply Bayes' Theorem: COVID Test

Complete all 3 steps. Show every calculation.

Given Information:



5% have COVID

$$P(\text{COVID}) = 0.05$$



Test detects 90%

$$P(+ | \text{COVID}) = 0.90$$



False positives 8%

$$P(+ | \text{No COVID}) = 0.08$$

? A person tests POSITIVE. What is $P(\text{COVID} | \text{positive test})$?

Step 1

List all known values: $P(\text{COVID}) = ___$, $P(\text{No COVID}) = 1 - ___ = ___$, $P(+ | \text{COVID}) = ___$, $P(+ | \text{No COVID}) = ___$

Step 2

Calculate $P(+)$ using Total Probability: $P(+) = (P(+ | \text{COVID}) \times P(\text{COVID})) + (P(+ | \text{No COVID}) \times P(\text{No COVID})) = (___ \times ___) + (___ \times ___) = ___$

Step 3

Apply Bayes and Interpret: $P(\text{COVID} | +) = (P(+ | \text{COVID}) \times P(\text{COVID})) / P(+) = ___ / ___ \approx ___$ Interpretation: what should the patient do next?

Activity Solution – COVID Bayes Calculation

✓ Model Answers

Step 1

Known Values →

$$\begin{array}{lcl} P(\text{COVID}) = 0.05 & | & P(\text{No COVID}) = 1 - 0.05 = 0.95 \\ P(\text{COVID} | +) = 0.90 & | & P(+ | \text{No COVID}) = 0.08 \end{array}$$

Step 2

Total Probability $P(+)$ →

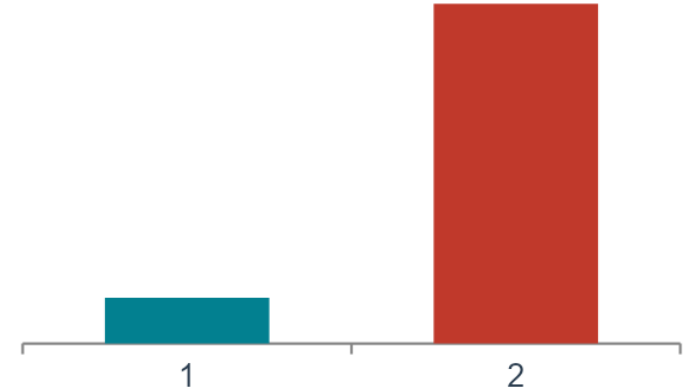
$$\begin{aligned} P(+) &= (0.90 \times 0.05) + (0.08 \times 0.95) \\ &= 0.045 + 0.076 \\ &= 0.121 \end{aligned}$$

Step 3

Bayes' Formula →

$$\begin{aligned} P(\text{COVID} | +) &= (0.90 \times 0.05) / 0.121 \\ &= 0.045 / 0.121 \\ &\approx 0.372 = 37.2\% \end{aligned}$$

$P(\text{COVID})$ Before vs After +ve Test



Interpretation: Despite testing POSITIVE, only 37.2% chance of having COVID. Why?

Because 95% of people tested DON'T have it, and the 8% false positive rate creates many false alarms. The low base rate dominates. Recommendation: Confirm with a SECOND test — the 37.2% posterior becomes the new prior, and two positive tests would yield ~92%+.

COVID Test — Full 3-Step Bayes Calculation

Given: $P(\text{COVID})=0.05$, $P(+|\text{COVID})=0.90$, $P(+|\text{No COVID})=0.08$

Step 1 — List All Known Values:

$P(\text{COVID}) = 0.05$ (5% of people have COVID)

$P(\text{No COVID}) = 1 - 0.05 = 0.95$ (95% are healthy)

$P(+|\text{COVID}) = 0.90$ (test detects 90% of sick people)

$P(+|\text{No COVID}) = 0.08$ (8% false positive on healthy people)

Step 2 — Calculate $P(+)$ using Total Probability:

$P(+) = (P(+|\text{COVID}) \times P(\text{COVID})) + (P(+|\text{No COVID}) \times P(\text{No COVID}))$

$P(+) = (0.90 \times 0.05) + (0.08 \times 0.95)$

$P(+) = 0.045 + 0.076 = 0.121$

Meaning: 12.1% of ALL tested people will test positive (sick OR false alarm).

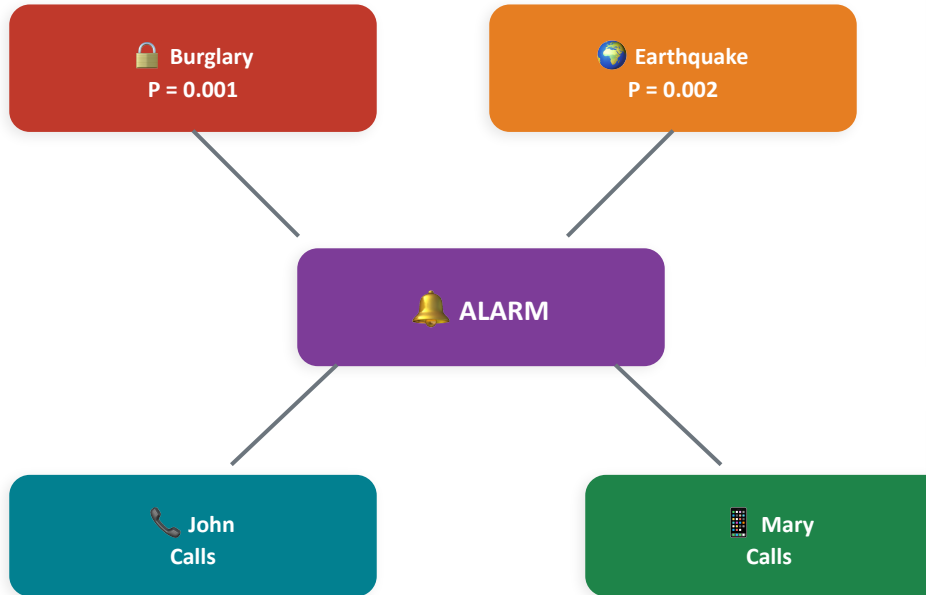
Step 3 — Apply Bayes Formula:

$P(\text{COVID}|+) = (0.90 \times 0.05) / 0.121 = 0.045 / 0.121 = 37.2\%$

Despite testing positive, only 37.2% chance of having COVID! Why higher than disease example (8.76%)? COVID affects 5% vs 1% — less rare means positives are more meaningful. Recommendation: Get a SECOND confirmatory test!

Bayesian Networks — Maps of Cause and Effect

When MANY causes affect MANY effects — a Bayesian Network shows them all with probabilities:



CPT — P(Alarm ON Burglary, Earthquake)		
Burglary	Earthquake	P(Alarm=ON)
YES	YES	0.95
YES	NO	0.94
NO	YES	0.29
NO	NO	0.001

Forward Inference

Cause → Effect: Burglary → P(Alarm) → P(JohnCalls) ≈ 85%

Backward Inference

Effect → Cause: John+Mary called → P(Burglary) rises 0.1% → 28.4%!

Bayesian Networks — How the Numbers Are Calculated

The network: Burglary(0.001) and Earthquake(0.002) can both trigger Alarm → which causes John and Mary to call.

CPT Table — Given by domain experts (not calculated):

Burglary YES + Earthquake YES → $P(\text{Alarm ON}) = 0.95$
Burglary YES + Earthquake NO → $P(\text{Alarm ON}) = 0.94$
Burglary NO + Earthquake YES → $P(\text{Alarm ON}) = 0.29$
Burglary NO + Earthquake NO → $P(\text{Alarm ON}) = 0.001$ (rare malfunction)

Forward Inference — How we get 85% for John Calls:

If Burglary → Alarm rings $\approx 94\%$ → John calls 90% of the time when alarm rings
 $P(\text{John calls} \mid \text{Burglary}) = 0.94 \times 0.90 = 85\%$

Backward Inference — How we get 28.4% after John AND Mary call:

Start: $P(\text{Burglary}) = 0.001$ (0.1%)
After John calls → apply Bayes → $P(\text{Burglary})$ rises to 1.67%
After Mary ALSO calls → 1.67% becomes new prior → apply Bayes again → $P(\text{Burglary}) = 28.4\%$
Each new evidence updates the belief! That is how Bayesian Networks work.

Quiz — Bayes' Theorem & Bayesian Networks



Scenario-Based Quiz — Think before checking answers!

Q1

Spam Filter: $P(\text{Spam})=0.4$. 'FREE' appears in 80% of spam and 5% of non-spam. A new email contains 'FREE'. Calculate $P(\text{Spam} \mid \text{'FREE'})$. Show all 3 steps. What does Gmail do with hundreds of such word probabilities?

Q2

Bayesian Network — Direction: In the Burglar Alarm network, John calls. (a) Is this forward or backward inference? (b) Before any call, $P(\text{Burglary})=0.001$. After John's call, does $P(\text{Burglary})$ increase or decrease? Why?

Q3

Base Rate Effect: A disease affects 0.1% of people. Test is 99% accurate for sick and 1% false positives. Without calculating, explain in plain English WHY a positive test result still means you probably DON'T have the disease.

Quiz Answers — Bayes' Theorem & Bayesian Networks



Model Answers

A1

$P(\text{FREE}) = (0.80 \times 0.40) + (0.05 \times 0.60) = 0.32 + 0.03 = 0.35$. $P(\text{Spam} | \text{FREE}) = (0.80 \times 0.40) / 0.35 = 0.32 / 0.35 \approx 91.4\%$. Gmail combines hundreds of word probabilities multiplicatively — each word updates the posterior which becomes the new prior for the next word, achieving ~99.9% accuracy.

A2

(a) BACKWARD (diagnostic) inference — we observe an effect (call) and reason back to the cause (burglary). (b) $P(\text{Burglary})$ INCREASES. John calling is evidence the alarm was on; the most likely cause of the alarm firing is a burglary. The evidence makes burglary more plausible despite the very low prior.

A3

Even with a 99% accurate test, when the disease affects only 0.1% of people (1 in 1000), the vast majority being tested are HEALTHY. The 1% false positive rate applied to that large healthy population generates many false alarms — far outnumbering the true positives from the tiny sick group. The base rate (how rare the disease is) dominates and overwhelms even a very accurate test.

Spam Filter — Gmail Bayes Calculation Step by Step

Scenario: $P(\text{Spam})=0.4$, $P(\text{FREE}|\text{Spam})=0.80$, $P(\text{FREE}|\text{Not Spam})=0.05$

Step 1 — List Known Values:

$P(\text{Spam}) = 0.40$, $P(\text{Not Spam}) = 1 - 0.40 = 0.60$

Step 2 — Total Probability $P(\text{FREE})$:

$P(\text{FREE}) = (0.80 \times 0.40) + (0.05 \times 0.60)$

$P(\text{FREE}) = 0.32 + 0.03 = 0.35$

Meaning: 35% of ALL emails contain the word FREE (spam and non-spam combined).

Step 3 — Apply Bayes:

$P(\text{Spam}|\text{FREE}) = (0.80 \times 0.40) / 0.35 = 0.32 / 0.35 = 91.4\%$

An email containing FREE has a 91.4% chance of being spam!

How Gmail Reaches 99.9% Accuracy:

Gmail does not just check ONE word. It checks hundreds — FREE, WINNER, CLICK HERE, URGENT, CONGRATULATIONS etc.

For each word, the posterior becomes the new prior for the next word check.

Example: After FREE → 91.4% spam. After WINNER also found → updates to even higher. After CLICK HERE → higher still. After 100+ words → 99.9% accuracy!

PART 5

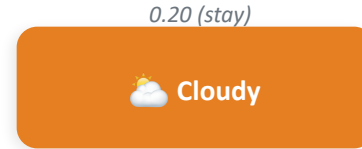
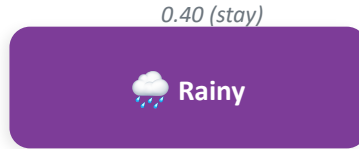
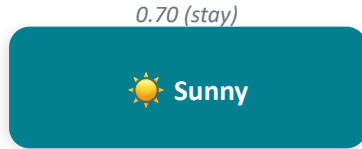
Markov Models & Fuzzy Logic

Predicting sequences • Reasoning with degrees of truth

Markov Chains — The Present Is All You Need

Markov Property: the next state depends *ONLY* on the current state — not on history.

 Board Game: Your token is on square 34. 5 moves or 50 moves to get there? *DOESN'T MATTER*. Next position = square 34 + dice roll only. History is irrelevant — that's Markov Property!



From \ To	 Sunny	 Rainy	 Cloudy
 Sunny	0.70	0.20	0.10
 Rainy	0.30	0.40	0.30
 Cloudy	0.50	0.30	0.20

Every ROW sums to 1.0 — must go SOMEWHERE

Worked: Rainy today → P(Sunny in 2 days)	
Rainy→Sunny→Sunny	$0.30 \times 0.70 = 0.21$
Rainy→Rainy→Sunny	$0.40 \times 0.30 = 0.12$
Rainy→Cloudy→Sunny	$0.30 \times 0.50 = 0.15$
Total	$0.21 + 0.12 + 0.15 = 0.48$

Markov Chain — The 3-Path Calculation Fully Explained

Question: Today is RAINY. What is $P(\text{Sunny in 2 days})$?

Why are there 3 paths? In 2 steps, the weather on DAY 1 (tomorrow) can be Sunny, Rainy, or Cloudy — these are 3 different routes to reach Sunny on DAY 2.

Path 1: Rainy → Sunny → Sunny

$$P = P(\text{Rainy} \rightarrow \text{Sunny}) \times P(\text{Sunny} \rightarrow \text{Sunny}) = 0.30 \times 0.70 = 0.21$$

Path 2: Rainy → Rainy → Sunny

$$P = P(\text{Rainy} \rightarrow \text{Rainy}) \times P(\text{Rainy} \rightarrow \text{Sunny}) = 0.40 \times 0.30 = 0.12$$

Path 3: Rainy → Cloudy → Sunny

$$P = P(\text{Rainy} \rightarrow \text{Cloudy}) \times P(\text{Cloudy} \rightarrow \text{Sunny}) = 0.30 \times 0.50 = 0.15$$

Final Answer = $0.21 + 0.12 + 0.15 = 0.48$ = 48% chance of Sunny in 2 days

WHY multiply WITHIN a path? Both steps must happen in sequence — like needing a key AND a door to open a room.

WHY add ACROSS paths? Any one of the 3 routes can get you to Sunny — like counting all roads from Lahore to Islamabad.

Markov Property: Note that HISTORY does not matter — only 'Rainy today' was needed to calculate this.

Hidden Markov Models — When States Are Invisible

HMM: hidden states you cannot see directly — only observations (symptoms) reveal them:

Markov Chain

States are VISIBLE

You see: Sunny → Rainy → Cloudy
(direct observation of state)

HMM

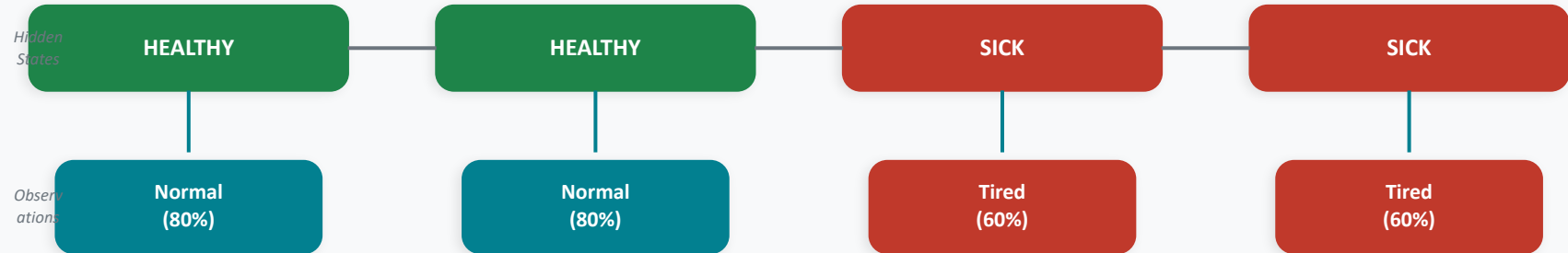
States are HIDDEN 🧟

Cannot see the state directly

Can see: Observations (symptoms)

Hidden state CAUSES observations

HMM Example — Health Monitoring



Speech Recognition: Hidden States = phonemes (/p/,/æ/,...), Observations = audio frequencies. HMM maps audio → phonemes → words!

Hidden Markov Models — Two Types of Probabilities Explained

Key Difference from Markov Chain: In HMM, you CANNOT see the state. You only see observations (symptoms/clues).

Transition Probabilities — How hidden states change:

Healthy → Healthy: 0.70 (70% chance of staying healthy tomorrow)

Healthy → Sick: 0.30 (30% chance of getting sick)

Sick → Sick: 0.60 (if sick, 60% chance of staying sick)

Sick → Healthy: 0.40 (40% chance of recovering)

Each row adds to 1.0 — must go somewhere!

Emission Probabilities — What each state produces as observations:

Healthy state produces: Normal feeling 80%, Tired feeling 20%

Sick state produces: Normal feeling 40%, Tired feeling 60%

The HMM Question it Answers:

Given observations: Tired, Tired, Normal, Tired — what is the most likely hidden sequence?

HMM calculates: probably Sick, Sick, Healthy, Sick.

Real World — Speech Recognition (Siri/Google):

Hidden States = phonemes (/p/, /ae/, /t/) — cannot directly see these

Observations = audio frequencies from microphone

HMM maps: audio frequencies → phonemes → WORDS. That is how voice assistants understand you!

Quiz — Markov Models

Q1

E-Commerce: States Browse, Cart, Checkout, Buy, Leave. From Cart: $P(\text{Checkout})=0.5$, $P(\text{Leave})=0.3$, $P(\text{Browse})=0.2$. From Checkout: $P(\text{Buy})=0.7$, $P(\text{Leave})=0.3$. Customer is in CART. What is $P(\text{Buy in exactly 2 steps})$? Show the multiplication.

Q2

Markov vs HMM: A psychologist tracks patient wellbeing. She CANNOT see if someone is 'Depressed' or 'Stable'. She observes: 'Engaged', 'Withdrawn', 'Tearful'. Is this a Markov Chain or an HMM? Identify the hidden states and observations.

Q3

Markov Property: To predict a chess player's NEXT move, do you need their full game history or just the current board position? Which AI principle justifies your answer? Why is history irrelevant?

Quiz Answers — Markov Models

Model Answers

A1

$P(\text{Cart} \rightarrow \text{Checkout} \rightarrow \text{Buy}) = P(\text{Checkout} | \text{Cart}) \times P(\text{Buy} | \text{Checkout}) = 0.5 \times 0.7 = 0.35 = 35\%$. Chain rule along the Markov path: multiply transition probabilities. Markov Property holds: from Cart, only current state matters — no need for history of how the customer arrived at Cart.

A2

HMM. Hidden states: Depressed / Stable (cannot observe directly). Observations: Engaged, Withdrawn, Tearful (visible behaviours). The hidden state CAUSES the observation probabilities. A Markov Chain would be wrong here because the state is unobservable — you cannot directly see someone's emotional state.

A3

Only the CURRENT board position — not the history of moves. Markov Property: the future depends only on the present state. The current board position already encodes the outcome of all previous moves — the history is embedded in the present. Past moves are irrelevant to evaluating the current position.

Activity — Markov Chain: Customer Journey

Calculate probabilities, then give a business recommendation.

Customer Journey States and Transition Probabilities



State	→ Cart	→ Checkout	→ Buy	→ Leave	→ Browse
Browse	0.30	—	—	0.50	0.20
Cart	—	0.50	—	0.30	0.20
Checkout	—	—	0.70	0.20	0.10

Q1 A customer is browsing. What is $P(\text{they reach Cart in 1 step})$?

Q2 What is $P(\text{Browse} \rightarrow \text{Cart} \rightarrow \text{Checkout})$ in exactly 2 steps?

Q3 Which state has the HIGHEST Leave probability? What UX change would you recommend?

Activity Solution – Customer Journey Markov Chain

✓ Model Answers

Q1 Answer

$P(\text{Cart} \mid \text{Browse}) = 0.30 = 30\%$. Read directly from the transition table, row 'Browse', column 'Cart'. Markov Property: only current state (Browse) needed — no history required.

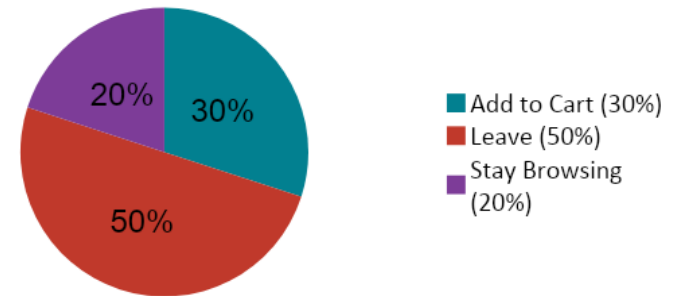
Q2 Answer

$P(\text{Browse} \rightarrow \text{Cart} \rightarrow \text{Checkout}) = P(\text{Cart} \mid \text{Browse}) \times P(\text{Checkout} \mid \text{Cart}) = 0.30 \times 0.50 = 0.15 = 15\%$. Multiply along the Markov path. Only 15% of browsing sessions convert to checkout in 2 steps.

Q3 Answer

BROWSE has the highest Leave probability: 50%! UX recommendations: better performance, faster load time. Reducing Browse→Leave from 50% to 30% could nearly double

From BROWSE state



Customer Journey — Full Calculation and Business Insight

Transition Table: Browse→Cart=0.30, Browse→Leave=0.50, Browse→Browse=0.20

Cart→Checkout=0.50, Cart→Leave=0.30, Cart→Browse=0.20

Checkout→Buy=0.70, Checkout→Leave=0.20, Checkout→Browse=0.10

Q1: P(Cart in 1 step from Browse) = ?

Answer: Read directly from table → $P(\text{Cart} | \text{Browse}) = 0.30 = 30\%$

Markov Property: Only need to know current state (Browse). No history needed.

Q2: P(Browse → Cart → Checkout) in exactly 2 steps = ?

Answer: Multiply along the path:

$P(\text{Cart} | \text{Browse}) \times P(\text{Checkout} | \text{Cart}) = 0.30 \times 0.50 = 0.15 = 15\%$

Only 15% of browsing sessions reach checkout in 2 steps. This is why UX matters!

Q3: Which state has highest Leave probability?

BROWSE has 50% Leave rate — HIGHEST of all states!

Cart = 30% leave, Checkout = 20% leave

UX Recommendations for Browse: Better search and categories, personalized recommendations, show discounts upfront, faster page loading.

Business Impact: Reducing Browse Leave from 50% to 30% could nearly DOUBLE the conversion rate!

Fuzzy Logic — Degrees of Truth, Not Just Yes/No

Classical logic = light switch (ON/OFF). Fuzzy logic = dimmer switch (any value 0 to 1):

❌ Classical Logic Problem

Cutoff: above 180cm = TALL.

Ali = 179.9cm → NOT TALL

Bilal = 180.1cm → TALL

A 0.2cm difference decides 'tall'?!
That feels completely WRONG.

✅ Fuzzy Logic Solution

Membership degrees (0.0 → 1.0):

160cm → 0.0 (not tall at all)

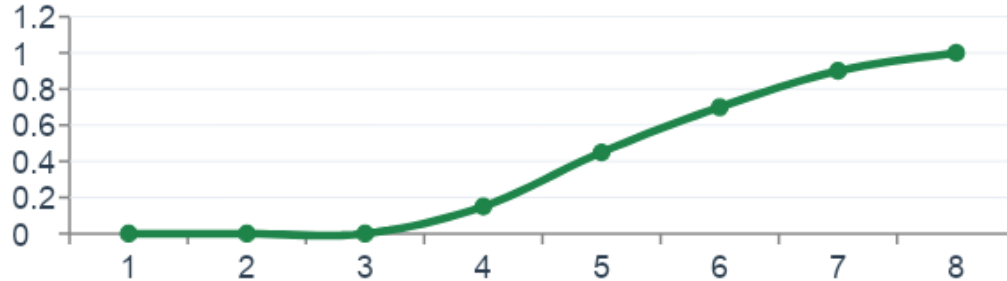
170cm → 0.3 (a bit tall)

178cm → 0.7 (quite tall)

185cm → 1.0 (fully tall)

Smooth gradient — no sharp cutoff!

Fuzzy Membership Function — TALL



Fuzzy Operations (Ali: tall=0.7, fast=0.4)

Fuzzy AND	$\text{MIN}(0.7, 0.4) = 0.4$	Weakest link
Fuzzy OR	$\text{MAX}(0.7, 0.4) = 0.7$	Strongest
Fuzzy NOT	$1 - 0.7 = 0.3$	Complement

Fuzzy Logic — AND, OR, NOT Calculations with Full Reasoning

Given: Ali has $\mu(\text{tall}) = 0.7$ and $\mu(\text{fast}) = 0.4$

Classical Logic Problem First:

If cutoff is 180cm $\rightarrow$ Ali at 179.9cm = NOT TALL, Bilal at 180.1cm = TALL. A 0.2cm difference decides everything — feels completely wrong!

Fuzzy solution: Ali at 179.9cm gets $\mu(\text{tall}) = 0.85$ — almost tall but not fully. Much more natural!

Fuzzy AND = $\text{MIN}(\mu_1, \mu_2)$:

$\mu(\text{tall AND fast}) = \text{MIN}(0.7, 0.4) = 0.4$

WHY MIN? AND means BOTH must be true. Chain is only as strong as its weakest link. Ali is limited by fast (0.4), not tall (0.7).

Plain English: Ali is 40% both tall and fast.

Fuzzy OR = $\text{MAX}(\mu_1, \mu_2)$:

$\mu(\text{tall OR fast}) = \text{MAX}(0.7, 0.4) = 0.7$

WHY MAX? OR means AT LEAST ONE true. Take the strongest. Ali is saved by being 70% tall.

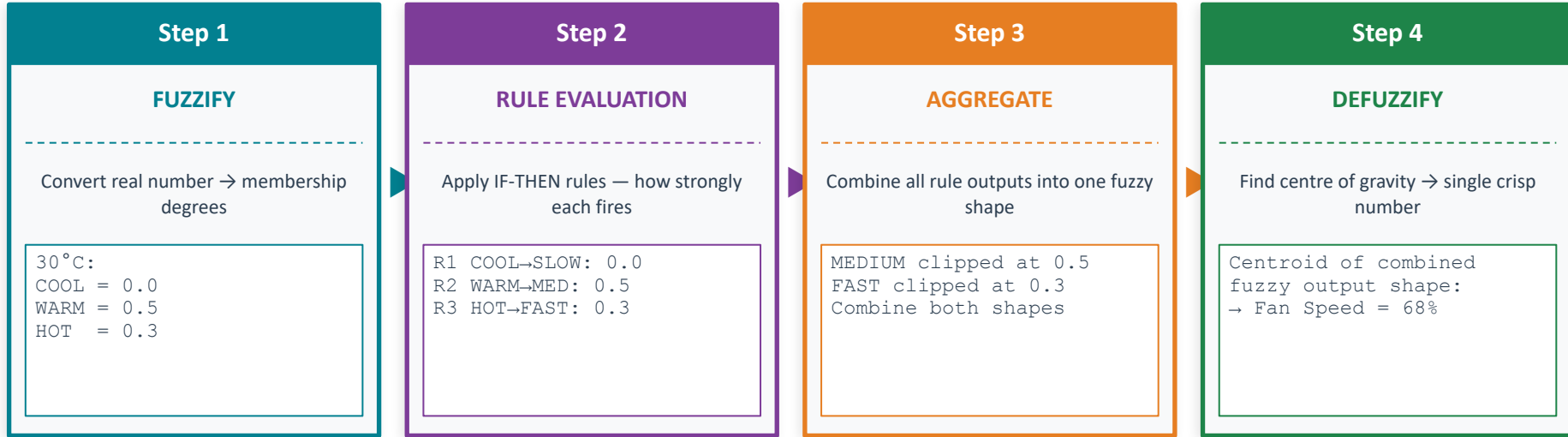
Fuzzy NOT = $1 - \mu$:

$\mu(\text{NOT tall}) = 1 - 0.7 = 0.3$

WHY 1 minus? If 70% tall then must be 30% not-tall. They must add to 1.0 (100%).

Fuzzy Inference System — 4 Steps with AC Controller Example

Input 30°C → Fuzzify → Apply Rules → Aggregate → Defuzzify → Output: 68% fan speed



AC Controller Fuzzy Rules:

Rule	IF Temperature is...	THEN Fan Speed is...	Fires at 30°C
1	COOL (below ~25°C)	SLOW	0.0 — does not fire
2	WARM (22°C–32°C)	MEDIUM	0.5 — fires half strength
3	HOT (above 30°C)	FAST	0.3 — fires some strength

Fuzzy Inference System — How 68% Fan Speed is Really Calculated

Input: 30 degrees Celsius. **Output:** Fan speed percentage.

Step 1 — FUZZIFY (convert 30C to membership degrees):

$\mu(\text{COOL}) = 0.0$ (30C is not cool at all — COOL peaks around 20C)

$\mu(\text{WARM}) = 0.5$ (30C is halfway into the WARM range of 22C to 32C)

$\mu(\text{HOT}) = 0.3$ (30C is entering HOT territory — HOT peaks at 35C)

Step 2 — RULE EVALUATION (rules fire at their membership strength):

R1: IF COOL → SLOW fan: fires at 0.0 (does NOT fire)

R2: IF WARM → MEDIUM fan: fires at 0.5 (half strength)

R3: IF HOT → FAST fan: fires at 0.3 (some strength)

Step 3 — AGGREGATE (clip and combine output shapes):

MEDIUM fan speed triangle is clipped at 0.5 (top of triangle cut off)

FAST fan speed triangle is clipped at 0.3

Combine both clipped shapes into one combined fuzzy region.

Step 4 — DEFUZZIFY using Centroid Method:

Weighted average: each fan speed value weighted by its membership degree.

Fan speeds near MEDIUM weighted at 0.5, near FAST weighted at 0.3.

The balance point (center of gravity) of combined shape = approximately 68% fan speed.

More precise sampling of points → closer to exactly 68%. This is real math, not an assumption!

Quiz — Fuzzy Logic

Think before checking answers!

Q1

AC at 22°C: $\mu_{\text{COOL}}(22)=0.8$, $\mu_{\text{WARM}}(22)=0.2$, $\mu_{\text{HOT}}(22)=0$. Rules: IF COOL $\rightarrow$ SLOW, IF WARM $\rightarrow$ MEDIUM, IF HOT $\rightarrow$ FAST. (a) Which rules fire and at what strength? (b) After defuzzification, what does the fan approximately do?

Q2

Fuzzy Operations — Patient X: $\mu_{\text{young}}=0.6$, $\mu_{\text{healthy}}=0.8$. Calculate: $\mu(\text{young AND healthy})$, $\mu(\text{young OR healthy})$, $\mu(\text{NOT young})$. Interpret each result in ONE plain English sentence.

Q3

Real World: Why does a modern anti-lock braking (ABS) system use Fuzzy Logic instead of classical on/off logic? What SPECIFIC advantage do membership degrees give compared to binary rules?

Quiz Answers — Fuzzy Logic

Model Answers

A1

(a) Rule 1 (COOL→SLOW) fires at 0.8. Rule 2 (WARM→MEDIUM) fires at 0.2. Rule 3 (HOT→FAST) fires at 0.0. (b) SLOW output clipped at 0.8 dominates; MEDIUM at 0.2. After defuzzification, fan runs at roughly 20–30% speed — mostly slow with a slight medium component. Room is cool, AC barely needs to run.

A2

$\mu(\text{young AND healthy}) = \min(0.6, 0.8) = 0.6$ — Patient is 60% both young and healthy (limited by youth). $\mu(\text{young OR healthy}) = \max(0.6, 0.8) = 0.8$ — Patient is 80% young or healthy (best of the two). $\mu(\text{NOT young}) = 1 - 0.6 = 0.4$ — Patient is 40% not young (middle-aged).

A3

Classical ABS: binary 'IF locked THEN release' → jerky on-off braking. Fuzzy ABS: graduated rules like 'IF wheel_slip is HIGH AND speed is FAST THEN reduce_braking SIGNIFICANTLY' — applied 14+ times per second. Membership degrees allow smooth, proportional pressure adjustments, giving shorter stopping distances and maintaining steering control throughout the braking event.

Fuzzy Logic Quiz Solutions

Q1: AC at 22C — $\mu(\text{COOL})=0.8$, $\mu(\text{WARM})=0.2$, $\mu(\text{HOT})=0.0$

Rule 1 (COOL→SLOW) fires at strength 0.8 — dominates strongly

Rule 2 (WARM→MEDIUM) fires at strength 0.2 — very weak

Rule 3 (HOT→FAST) fires at strength 0.0 — completely silent

SLOW output clipped at 0.8 dominates, MEDIUM at 0.2 is minor.

After defuzzification: fan runs at roughly 20-30% speed. Room is cool, AC barely runs.

Q2: Patient X — $\mu(\text{young})=0.6$, $\mu(\text{healthy})=0.8$

AND: $\text{MIN}(0.6, 0.8) = 0.6 \rightarrow 60\%$ both young and healthy (limited by youth)

OR: $\text{MAX}(0.6, 0.8) = 0.8 \rightarrow 80\%$ young or healthy (saved by good health)

NOT young: $1 - 0.6 = 0.4 \rightarrow 40\%$ not-young (somewhat middle-aged)

Q3: Why does ABS braking use Fuzzy Logic?

Classical ABS: IF wheel locked THEN release brake fully $\rightarrow$ jerky, uncomfortable, longer stop

Fuzzy ABS: IF wheel_slip is HIGH AND speed is FAST THEN reduce_braking SIGNIFICANTLY

Applied 14 times per second with smooth, proportional pressure adjustments.

Result: Shorter stopping distances + full steering control maintained. Membership degrees allow graduated brake pressure, not just full-on or full-off!

Activity — Design a Fuzzy Cruise Control System

 **Pair Activity (12 min)** — Design the full fuzzy system. Apply it to a real scenario.

Setup: Input = Speed Error (current minus target speed). Negative = too slow. Zero = perfect. Positive = too fast. Output = Throttle adjustment (INCREASE, HOLD, DECREASE).

1. Define Fuzzy Sets for Speed Error

Name 3 sets (TOO_SLOW, ABOUT_RIGHT, TOO_FAST). For each, state where $\mu=1$ and where $\mu=0$.

2. Write 3 IF-THEN Rules

Connect speed error sets to throttle output sets. Make sure you have a rule for each speed error set.

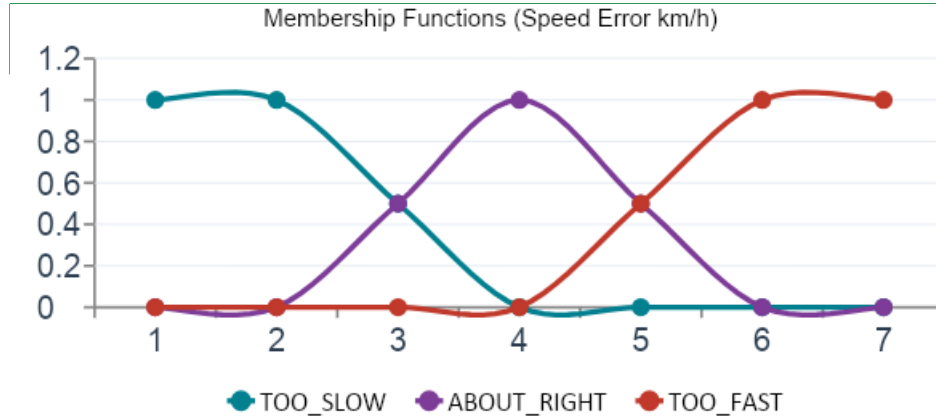
3. Fuzzify: Error = +5 km/h (too fast)

Calculate μ for each set. Use your ranges from Task 1. Which rules fire and at what strength?

4. Interpret and Extend

What should the throttle do for error=+5? What if error=-8 km/h (too slow)? Which rule dominates then?

Activity Solution – Fuzzy Cruise Control System



Rules

R1: IF TOO_SLOW → INCREASE throttle
R2: IF ABOUT_RIGHT → HOLD throttle
R3: IF TOO_FAST → DECREASE throttle

Rules Fire?

R2 fires at 0.5 → HOLD (0.5)
R3 fires at 0.5 → DECREASE (0.5)
R1 = 0.0 (no fire)

Error=+5 Fuzzify

$\mu_{\text{TOO_SLOW}}(+5) = 0.0$
 $\mu_{\text{ABOUT_RIGHT}}(+5) = 0.5$
 $\mu_{\text{TOO_FAST}}(+5) = 0.5$

Output

Both HOLD and DECREASE at 0.5. Defuzzify → slight throttle decrease. At -8

 *Real World Fuzzy: Rice cookers (Panasonic), camera autofocus, anti-lock brakes (ABS), air conditioning, washing machine load sensing, autopilot systems. Invented by Lotfi Zadeh (1965) — now inside billions of devices worldwide!*

Fuzzy Cruise Control

Setup: Speed Error = Current Speed - Target Speed

Fuzzy Sets: TOO_SLOW (peaks at -10), ABOUT_RIGHT (peaks at 0), TOO_FAST (peaks at +10)

Rules: R1: IF TOO_SLOW → INCREASE, R2: IF ABOUT_RIGHT → HOLD, R3: IF TOO_FAST → DECREASE

Scenario A: Error = +5 km/h (car is 5 km/h too fast)

Fuzzify: $\mu(\text{TOO_SLOW})=0.0$, $\mu(\text{ABOUT_RIGHT})=0.5$, $\mu(\text{TOO_FAST})=0.5$

Rule R1 fires at 0.0 → silent

Rule R2 fires at 0.5 → HOLD at half strength

Rule R3 fires at 0.5 → DECREASE at half strength

Both HOLD and DECREASE fire equally → After defuzzify: SLIGHT throttle decrease.

Car is only a little too fast (+5) → gentle correction, not a sudden brake!

Scenario B: Error = -8 km/h (car is 8 km/h too slow)

Fuzzify: $\mu(\text{TOO_SLOW})=0.8$, $\mu(\text{ABOUT_RIGHT})=0.2$, $\mu(\text{TOO_FAST})=0.0$

Rule R1 fires at 0.8 → INCREASE dominates strongly

Rule R2 fires at 0.2 → HOLD is very weak

After defuzzify: STRONG throttle increase. Car is quite slow → system pushes hard!

KEY INSIGHT: The further from target speed, the STRONGER the correction. Closer = gentler. Just like a skilled human driver naturally adjusts throttle smoothly!

The Big Picture — How It All Connects

TIER 1 — STORING KNOWLEDGE (What do we know?)

RDF

Facts as triples
Subject→Pred→Object

OWL

Rules & hierarchies
Automatic reasoning

SPARQL

Query patterns
SELECT/ASK/etc.

▼ What happens when facts are UNCERTAIN or INCOMPLETE?

TIER 2 — REASONING UNDER UNCERTAINTY (What might be true?)

Probability

P(event)
Scale 0→1

Bayes' Theorem

Update beliefs
with evidence

Bayesian Networks

Cause→Effect
maps+probs

Markov Models

Predict next
state from now

Fuzzy Logic

Degrees of
truth 0→1

Quick Selector: vague words → Fuzzy | sequences → Markov | causes+effects → BayesNet | updating beliefs → Bayes | rules → OWL | querying → SPARQL | storing facts → RDF